

Modular Iron-Bipyridine based Conjugated Aerogels as Catalysts for Oxygen Reduction Reaction

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Aerogels have a large surface area and a porous structure which makes them an attractive catalyst support material for fuel cells. Modifying the aerogels' building blocks and introducing catalytic sites into their backbones allows them to function as both catalysts and supports, thereby increasing the density and distribution of catalyst active sites. In this work, we studied conjugated aerogels containing iron-bipyridine catalytic sites for oxygen reduction reaction. To demonstrate the variation in physical and electrochemical properties of these aerogels, a series of aerogels were synthesized by Glaser coupling reaction. Iron salt was added to the three-dimensional gel to produce iron-bipyridine complexes and obtain atomically dispersed catalytic sites. The electrocatalytic activity and electrical conductivity of the aerogels were increased after their heat treatment to yield Fe-doped carbon aerogels. The control of metal loadings was accomplished by changing the bipyridine content during the aerogel synthesis, resulting in 0.9 - 4.1 %wt Fe. The results show a correlation between Fe loading, aerogel structure, and catalytic activity, achieving high performance for the oxygen reduction reaction in both acidic and alkaline media.

Introduction

Climate change and its implications call for changing the way we produce and use energy, and relying more on sustainable energy sources. This increases the motivation for clean energy technologies.^{1, 2} In the transition to sustainable energy, there is a great need for large energy storage. One of the most prominent solutions is the Hydrogen Economy,³ which is based on the production of green hydrogen from sustainable sources such as wind and sun. This is used in a wide variety of applications, from the metal industry to transportation and electricity production. At the heart of this relatively new scheme is fuel cell technology, which converts the chemical energy stored in hydrogen directly into electrical energy.⁴⁻⁶ One of the major impediments in the implementation of the Hydrogen Economy,⁷ is the high costs associated with its technologies, which also include fuel cells' costs.⁸

The sluggish kinetics of the oxygen reduction reaction (ORR) in FCs has the single largest impact on their performance. Many studies have been conducted to develop catalysts, improve the ORR thermodynamic efficiency, and increase the reaction's kinetics.⁹ In addition, reducing the cost of the catalysts, mostly based on platinum group metals (PGM), is also a significant challenge.^{10, 11} These issues are addressed by the continuous effort

to develop catalysts and cathode platforms based on PGM-free, earth-abundant materials.¹²⁻¹⁴

PGM-free ORR catalysts based on first-row transition metals, have shown good catalytic activity, where the state-of-the-art are atomically dispersed metal sites supported by nitrogen atoms embedded in carbon-based platforms (also referred to as M-N-Cs).¹⁵⁻¹⁷ These catalysts are produced in many variations such as molecular catalysts, containing nitrogen-rich molecules, and metal ions absorbed on carbon supports.¹⁸

Covalent organic frameworks (COF) in the form of aerogels offer many merits which make them very interesting platforms for PGM-free ORR catalysts.^{19, 20} By tailoring their building blocks, instead of adding the catalyst onto the support, the catalyst can be embedded into the support synthetically.²¹ This allows to improve control over both the chemical structure and the active site density, increasing metal utilization and improve mass transport.²²⁻²⁶ In addition, the physical and chemical properties can be tuned to optimize these aerogels for best performance in FCs. These highly porous hierarchical structures are usually heat treated to improve their electronic conductivity,^{27, 28} to be used as catalyst supports.

In our previous work we presented a unique design of conjugated aerogels containing Cu-bipyridine (bpy) as catalytic sites which were

integrated by an in-situ polymerization and complexation method.²² Control over the site density of ORR catalytic sites was achieved by changing the post-polymerization conditions.

Herein, we present control over site density during polymerization, preparing a series of aerogels enabled by the simple and modular synthesis. By increasing the bpy units' loading, we improved the chemical and physical structure of the conjugated aerogel, including an increase in the number of the Fe-bpy catalytic sites. We also report improved activity, by replacing the copper metal center with iron, which is generally known for its superior performance in ORR electrocatalysis (in the category of PGM-free) and for promoting ORR in the efficient four-electron reaction pathway.^{29, 30} This advanced version of bpy-based aerogels enables increasing the active site density and overall catalytic activity in both acidic and alkaline conditions.

Results and discussion

Poly(triethynylbenzen-bipyridine) (PTEBbpy) aerogels were synthesized by co-polymerization of 1,3,5- triethynyl benzene (TEB) and 5,5'-diethynyl-2,2'- bipyridine (DEbpy), by Glaser coupling reaction, using CuCl as a catalyst, as illustrated in **Figure 1**.²² Controlling the bpy sites was performed by changing the TEB:DEbpy monomer's ratio. The control over the aerogel composition is demonstrated by organic elemental analysis. **Table 1** shows the C, H, and N weight percent (%wt) of the experimental and expected theoretical values. The experimental nitrogen %wt fits the expected value very well and confirms the successful Glaser coupling reaction and the formation of the aerogel, which can also be seen at the bottom of **Figure 1**. The hydrogen %wt found to be higher than expected, probably due to unreacted ends or humidity.

To make the gel catalytically active, iron complexation was performed by adding iron nitrate solution in ethanol. After washing and supercritical drying, a PTEBbpyFe aerogel was obtained. Inductively coupled plasma mass spectrometry (ICP) measurements confirmed that iron was inserted successfully (**Table 2**).

The PTEBbpyFe aerogels (AG) were heat treated (HT) at 800°C under argon to improve the electrical conductivity and catalytic activity and produce carbon aerogels (PTEBbpyFe-HT). This temperature was selected carefully after studying the effect of

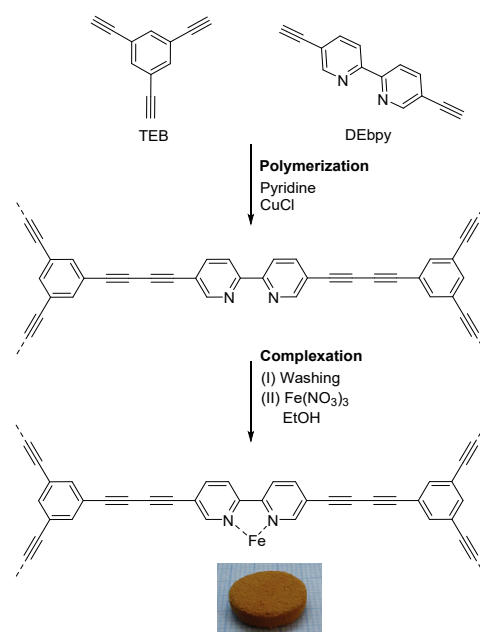


Figure 1. Scheme of the synthesis of the PTEBbpyFe aerogels, and the resulting aerogel after supercritical drying.

Table 1. CHN elemental analysis of PTEBbpy aerogels of different TEB: DEbpy ratio.

TEB:DEbpy ratio	Normalized [%wt]			Theory [%wt]		
	N	C	H	N	C	H
4:1	3.6	92.6	3.8	3.5	94.2	2.3
3:1	4.5	91.5	4	4.4	93.3	2.3
2:1	5.3	90.8	3.9	5.6	91.9	2.4
1:1	8.0	88.1	3.8	8.0	89.4	2.6
0.33:1	10.8	85.4	3.9	11.1	86.0	2.8

Table 2. Fe and Cu weight percent in PTEB_{0.33}bpyFe aerogels.

Aerogel	Fe [% wt]	Cu [% wt]
AG	4.45	2.62
HT	4.37	3.29
HT2	4.10	0.25

heat treatment on the conductivity and overall performance of aerogels in FCs in a previous work.³¹

To eliminate any elemental and unstable metal species that may have formed during the HT, acidic washes (AW) were performed according to a known procedure,³² producing PTEBbpyFe-AW aerogels. These changes were studied using X-ray diffraction (XRD) spectroscopy (**Figure 2**). After the HT, the Cu and Fe XRD patterns were observed at 43.4°, and 50.5° and 74.3° and 44.7°, for the Cu and Fe, respectively.³³⁻³⁵ These patterns were eliminated after the AW. It is well documented in the literature

that a second heat treatment can further increase the activity of PGM-free ORR catalysts,³² and thus it was performed here as well. The Fe and Cu particles observed in the XRD of the HT aerogel before the acid wash, were not observed after the second HT (PTEBbpyFe-HT2). This was also detected by scanning transmission electron microscopy (STEM-Figure S1 in SI).

ICP measurements were performed to study the change in the metal loadings after these treatments. The ICP results show a slight reduction in the Fe loading after the second heat treatment (HT2), from 4.45 to 4.10 %wt, whereas the Cu loading was significantly reduced from 2.62 to 0.25 %wt (**Table 2**). This low Cu loading, and the fact that most Fe-based PGM-free catalysts are considered to be much more ORR active than Cu-based ones, allows to neglect the contribution of the Cu sites to the ORR electrocatalysis.

The Fe loading could be tuned by changing the PTEB:bpy ratio in the aerogel. Increasing the bipyridine content increased the iron content as expected (**Figure 3**). An N:Fe ratio of about 5:1 was obtained for the three aerogels with higher bipyridine content (PTEB_{0.33}bpyFe-HT2, PTEB₁bpyFe-HT2 and PTEB₂bpyFe-HT2). Aerogels with lower bipyridine content showed higher N:Fe ratios of 7:1 and 11:1 for PTEB₃bpyFe-HT2 and PTEB₄bpyFe-HT2, respectively.

The coordination of Fe in the aerogel and the speciation of the nitrogen were studied using XPS before and after heat treatment and acid wash. As summarized in **Table 3** and shown in **Figure 4A-C**, the non-treated organic PTEBbpyFe aerogel contains only three distinct forms of nitrogen. The backbone characteristic pyridinic N (398.7 eV), N-Fe (399.7 eV) of the formed bpy-Fe complexes, and nitrate N (406.2 eV)³⁶, which most probably left over from the iron nitrate salt. After the first HT, graphitic N sites (401.0, 402.5 eV)³⁷ appeared in conjunction with the decrease in pyridinic N and N-Fe sites. After the second HT, a further increase in

Table 3. N1s binding energies, measured by XPS.

	Peak BE (eV)	Peak area [%]		
		AG	HT	HT2
Pyridinic	398.4-398.8	47.93	32.79	27.70
N-Fe	399.5-399.9	39.65	20.36	20.73
Graphitic	400.9-401.0	-	35.28	39.35
	402.3-402.5	-		
N-O and Nitrate	404.0-404.2	12.42	11.59	12.22
	406.0-406.5			

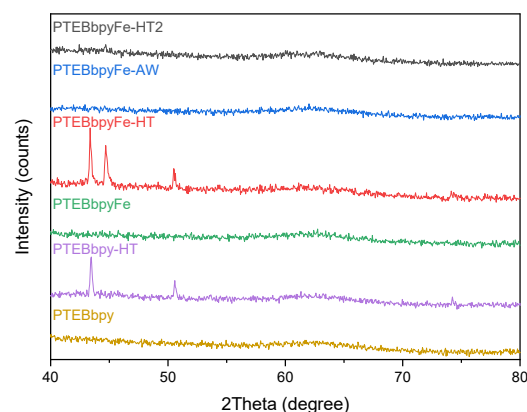


Figure 2. (A) XRD analysis of PTEB_{0.33}bpyFe aerogel under different treatments compared to PTEBbpy.

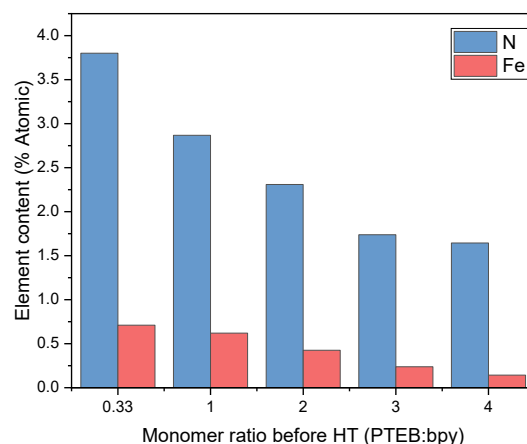


Figure 3. N and Fe percentage in PTEB_xbpyFe-HT2 aerogels as a function of initial monomer ratio, according to organic EA and ICP respectively.

graphitization has occurred on the account of pyridinic N. The consistency in the peaks' binding energies before and after HT indicates that the pyridinic nature of the Fe-N-C was retained.³⁸ It is also noticeable that the oxidized N sites' percentage was not affected by the HT processes (**Table 3**). To confirm the presence of N-Fe complex the aerogels were compared to DEBpy monomer which showed only pyridinic nitrogen (Figure S2 in SI).

Figure 5C-D shows the graphitization of the aerogels observed by annular dark-field scanning transmission electron microscopy (ADF-STEM), and confirmed by XRD and Raman spectroscopy (Figure S3 in SI). In addition, these images show the atomically dispersed iron atoms homogeneously distributed in the aerogel (examples are highlighted in orange circles in the figures). Secondary electron (SE) and lower magnification ADF-STEM images (**Figure 5A-B**) also show the aerogel's porous hierarchical structure. This is composed of interconnected particles with large void volumes in between. This morphology is expected to allow

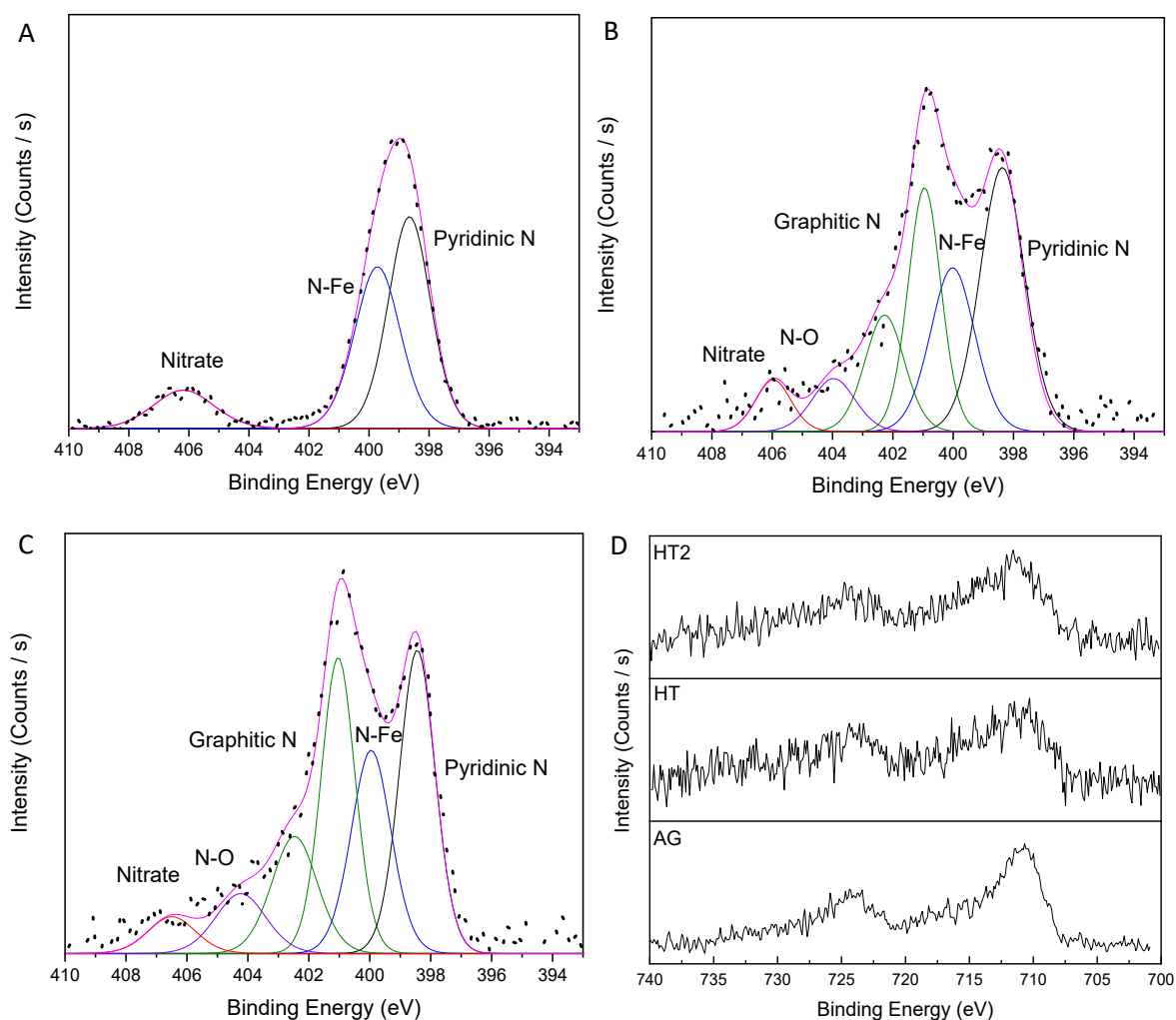


Figure 4. XPS spectra of PTEB_{0.33}bpyFe aerogel N 1s (A) before and (B) after HT (C) HT2 (D) Fe 2p spectra compare.

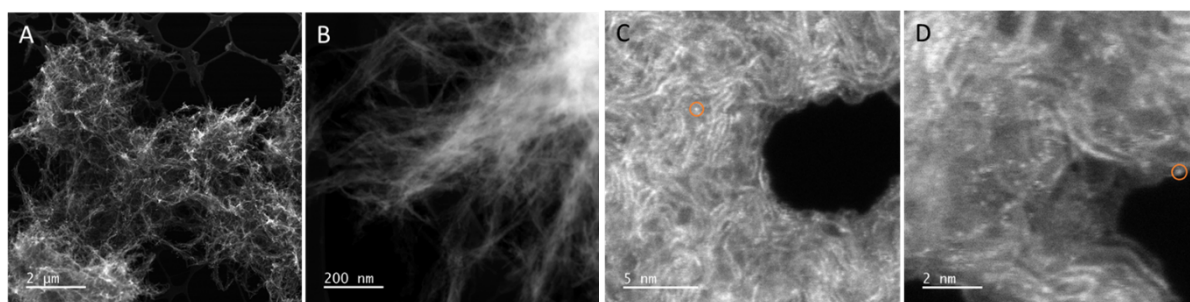


Figure 5. STEM images of PTEBbpyFe-HT2 obtained using (A) secondary electron and (B-D) annular dark field detectors.

efficient mass transport of oxygen to the active sites, and water out.

The aerogel's porous structure was analyzed by N₂ adsorption measurements, showing BET surface area of 568-668 m²/g (**Table 4**). According to the pore size distribution (by optimal DFT fitting) the aerogel samples can be divided into two groups. PTEB₃bpyFe-HT2 and PTEB₄bpyFe-HT2 consist of slit pores and have higher microporosity, whereas PTEB₂bpyFe-HT2, PTEB₁bpyFe-HT2, and PTEB_{0.33}bpyFe-HT2 are composed of both slit and

cylindric pores and offer a more mesoporous structure (**Figure 6**).

Table 4. PTEBbpyFe-HT2 aerogels BET surface area.

Aerogel	Surface Area [m ² /g]
PTEB _{0.33} bpyFe-HT2	599
PTEB ₁ bpyFe-HT2	568
PTEB ₂ bpyFe-HT2	607
PTEB ₃ bpyFe-HT2	668
PTEB ₄ bpyFe-HT2	640

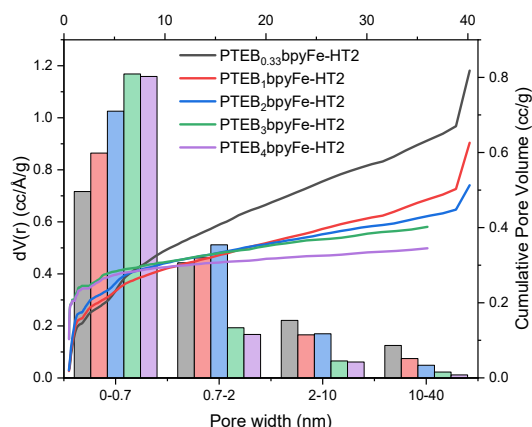


Figure 6. Pore size distribution and cumulative pore volume of PTEB_xbpyFe-HT2 aerogels, derived from N₂ adsorption isotherm.

Table 5. Fe loading measured using ICP vs. EASD from CV measurements of the PTEB_xbpyFe-HT2 aerogels.

Aerogel	ICP Fe loading (site/g)	EASD (site/g)
PTEB _{0.33} bpyFe-HT2	4.5×10^{20}	1.3×10^{19}
PTEB ₁ bpyFe-HT2	4.0×10^{20}	8.4×10^{19}
PTEB ₂ bpyFe-HT2	2.7×10^{20}	9.0×10^{19}
PTEB ₃ bpyFe-HT2	1.5×10^{20}	6.9×10^{19}
PTEB ₄ bpyFe-HT2	9.5×10^{19}	3.3×10^{19}

ORR Catalytic Activity

Cyclic voltammetry (CV) in acidic conditions (0.5 M H₂SO₄) demonstrates the role of N-Fe complexes as active sites. The CV of PTEBbpyFe-HT2 aerogel was compared to a metal-free PTEBbpy-HT2 aerogel (**Figure 7**). A quasi-reversible redox couple was observed after the addition of Fe to the aerogel, highlighting two significant features: (1) Fe was incorporated in the aerogel, and (2) it is electrochemically active ($E^\circ(\text{Fe}^{3+}/\text{Fe}^{2+}) = 0.63 \text{ V vs. RHE}$). Moreover, in the presence of oxygen, the reduction peak increases significantly. This hint that the Fe-N sites in the aerogel are indeed the ORR active sites, as was observed with other transition metal complexes such as porphyrins, corroles, and phthalocyanines before.³⁹⁻⁴²

The electrochemically active site density (EASD) of all PTEBbpyFe-HT2 aerogels was calculated by integration of the $\text{Fe}^{3+}/\text{Fe}^{2+}$ redox peaks from the CV under argon in acidic media and averaging the results (**Table 5**). As expected, the EASD is lower than the Fe loadings measured using ICP, but in contrast to most PGM-free ORR catalysts, it is within the same order of magnitude,^{43, 44} showing

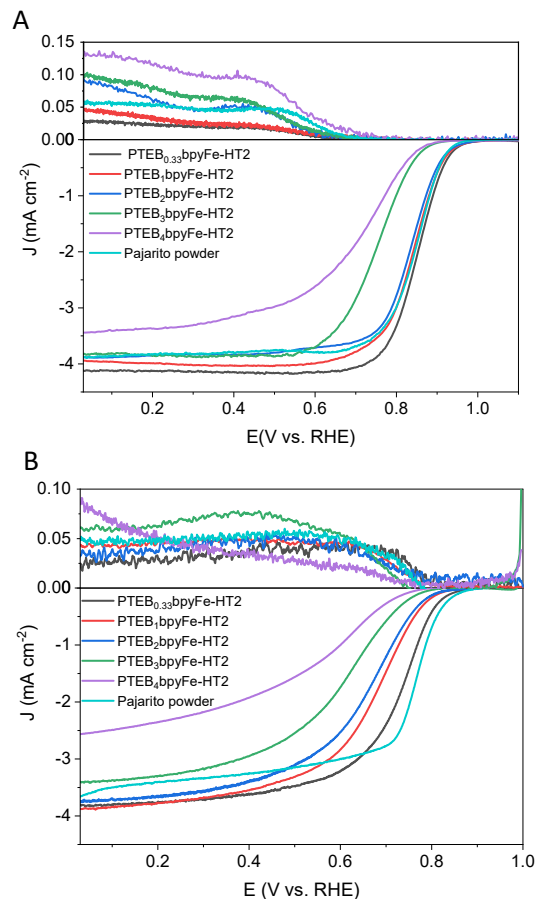


Figure 8. RRDE measurements at 900 rpm and 5 mV/s, of PTEB_xbpyFe-HT2 aerogels compared to commercial Pajarito powder, in O₂-saturated (A) 0.1 M KOH and (B) 0.5 M H₂SO₄ solutions. Catalyst loading 0.25 mg/cm².

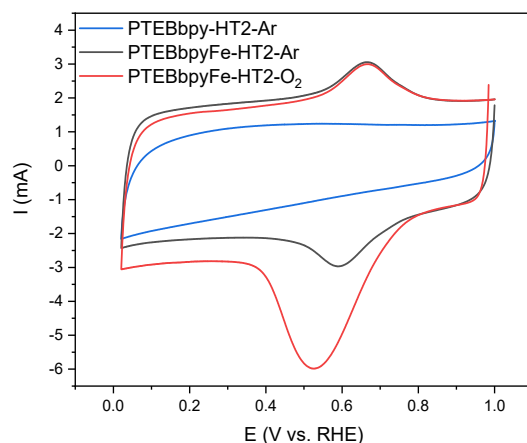
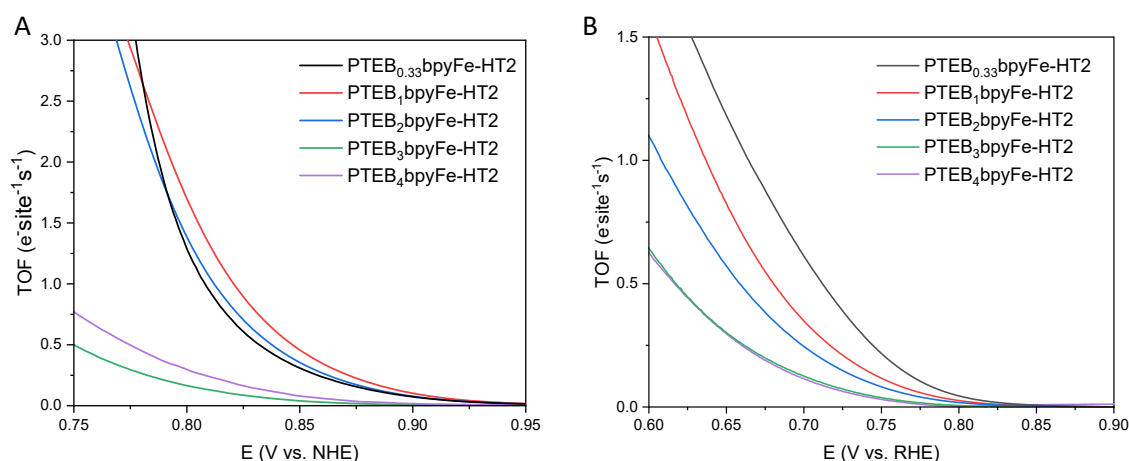


Figure 7. CV measurement of PTEBbpyFe-HT2 at 100 mV/s under Ar and O₂, and PTEBbpy-HT2 under Ar, in 0.5 M H₂SO₄. Catalyst loading 0.25 mg/cm².

that the hierarchical structure of the aerogel allows very good mass transport of electrolyte, and thus very good accessibility to the active site, which correlates with the pore structure: the aerogels

Table 6. ORR electrocatalytic performance of carbon aerogels in acidic and alkaline conditions.

Aerogel	Alkaline conditions (0.1 M KOH)				Acidic conditions (0.5 M H ₂ SO ₄)			
	E _{onset} ¹ [V vs. RHE]	Limiting current density [mA/cm ²] ²	Max. Peroxide yield ³ [%]	Electron number ⁴	E _{onset} ¹ [V vs. RHE]	Limiting current density [mA/cm ²] ²	Max. Peroxide yield ³ [%]	Electron number ⁴
PTEB _{0.33} bpyFe-HT2	1.02	-4.14	2.0	4.0	0.89	-3.80	4.8	3.8
PTEB ₁ bpyFe-HT2	1.01	-4.02	3.3	4.0	0.86	-3.84	4.3	3.9
PTEB ₂ bpyFe-HT2	1.00	-3.83	6.6	3.8	0.85	-3.69	5.1	3.7
PTEB ₃ bpyFe-HT2	0.94	-3.88	7.3	3.8	0.81	-3.37	7.4	3.4
PTEB ₄ bpyFe-HT2	0.85	-3.3	10.8	3.2	0.78	-2.49	10.0	2.5

¹E_{onset} was determined at 0.02 mA/cm².²Limiting current was determined at 0.30 V vs. RHE in alkaline, and 0.10V vs. RHE in Acid.³The collection efficiency of the ring from the disk was considered to be 37% of the peroxide yield, as provided by the manufacturer.**Figure 9.** TOF of PTEB_xbpyFe-HT2 aerogels in O₂-saturated (A) 0.1 M KOH and (B) 0.5 M H₂SO₄ solutions.

with the higher bpy content (PTEB₂bpyFe-HT2, PTEB₁bpyFe-HT2 and PTEB_{0.33}bpyFe-HT2) also have higher mesoporosity, which apparently allows better mass transport and thus increases the EASD, as was shown with different aerogel structures in the past.²¹

The combination of EASD with bipyridine content and pore structure manifests in the ORR electrocatalytic activity, measured using RRDE, in both alkaline and acidic conditions (**Figure 8**). Higher mesoporosity, bpy-Fe loading, and EASD resulted in higher onset potentials and higher limiting currents. This is also correlated with higher ORR selectivity for the four-electron reduction pathway, manifested by lower peroxide formation and higher electron number (**Table 6**). This trend is more noticeable in acidic environment, possibly due to slower ORR kinetics under these conditions.

In alkaline environment, PTEB₂bpyFe-HT2, PTEB₁bpyFe-HT2 and PTEB_{0.33}bpyFe-HT2 perform very similarly. Yet, the PTEB_{0.33}bpyFe-HT2, with the highest Fe loading and EASD, is the most effective in both alkaline and acidic solutions, with an onset potential of E_{onset} = 1.02 and 0.89 V vs. RHE, respectively.

Accordingly, higher turnover frequency (TOF) values were calculated for these three aerogels (**Figure 9**). As explained extensively in the work of P. Atanassov et al.⁴⁵ and cited works therein^{46, 47}, several different methods were proposed over the past decade for calculation of the active site density in order to calculate the TOF, in addition to some new methods developed since then for in situ in fuel cell measurements.^{43, 48-50} What seem to be the most straight-forward method nowadays is calculation of the EASD from integration of the

Fe³⁺/Fe²⁺ redox peaks. Hence, this method was used in this work to easily compare the results with other studies, although the EASD calculated this way may be overestimated according to some other measurement methods. The TOF was calculated considering the EASD and kinetic current (**Equation S4** in the SI), indicating the intrinsic activity of each aerogel. The TOF results are in good agreement with the electrochemical performance trend. Their variation is an indication that there is a difference in catalytic sites, which among others could be attributed to the synergistic effect between neighbouring active sites, which is promoted by their loading and the physical structure and porosity of the aerogels.

Summary and conclusions

This work presents the modularity of PTEBbpy aerogels by introducing Fe-bpy ORR active sites. By changing the monomers' ratio, we controlled both the aerogel's chemical and physical structure. This allowed us to alter the Fe-N catalytic site loading and electrochemical accessibility. The aerogels were thoroughly characterized and optimized to produce atomically dispersed Fe active sites in the aerogels. The changes in the aerogel's structure enabled the production of high range of catalytic performances. The most active catalyst produced contained the greatest amount of bipyridine units in the structure, producing high mesoporosity and allowing it to reach an iron loading of 4.1 %wt. This catalyst showed the highest EASD, TOF, and onset potential in both alkaline and acidic conditions.

Experimental section

Synthesis

Poly(triethynyl benzene bipyridine) were synthesized according to our previous work,²² with some changes. TEB (98+%, TCI) and DEbpy (96%, Chemieliva) (0.1 mmol together) and CuCl (97%, Sigma Aldrich) (0.1 mmol) were dissolved in 2 ml Pyridine using sonication (5-10 minutes). After stirring for 20 min at 40°C, the solution was left at room temperature (RT) for 24 hours, for gel formation and aging. Washing of the gels were performed using: pyridine, ethanol, 6 M HCl in ethanol and neutralization by 0.05 M KOH in ethanol: water (4:1 v/v), and transferred back to ethanol and finally extra dry ethanol before the gels were dried using supercritical CO₂ (Leica EM CPD300).

The TEB:DEbpy ratios were changed according to the desired product. For PTEB₄bpy: TEB (0.08 mmol,

4 equiv.), DEbpy (0.02 mmol, 1 equiv.). PTEB₃bpy: TEB (0.075 mmol, 3 equiv.), DEbpy (0.025 mmol, 1 equiv.). PTEB₂bpy: TEB (0.066 mmol, 2 equiv.), DEbpy (0.033 mmol, 1 equiv.). PTEB₁bpy: TEB (0.05 mmol, 1 equiv), DEbpy (0.05 mmol, 1 equiv.). PTEB_{0.33}bpy: TEB (0.025 mmol, 1 equiv.), DEbpy (0.075 mmol, 3 equiv.).

Then for metalation the gels were soaked in iron nitrate (Fe(NO₃)₃*9H₂O, 98+%, Thermo scientific, 1 equiv. in ethanol) for 3 days before additional washes with ethanol and supercritical drying, to obtain PTEBbpyFe aerogels.

For heat treatment, the aerogels were placed in a tube furnace (Thermo Scientific-Lindberg Blue M) and heated at a rate of 1.75°C/min up to 800°C under an argon atmosphere. The high temperature was maintained for 2 h before gradual cooling, to produce PTEBbpyFe-HT aerogels. Then the aerogels were soaked in 0.5 M H₂SO₄ in water at 80°C for 8 h, and washed with ethanol:water (2:1 v/v), and ethanol, then ambient dried to produce PTEBbpyFe-AW. An additional HT was performed at 800°C for 3 h, to obtain PTEBbpy-HT2 aerogels.

Characterization

The characterization of the aerogels was performed by XPS measurements using Nexsa spectrometer with a mono-chromated, micro-focused, low power Al K α X-ray source (photon energy 1486.6 eV at 72 W). Advantage software was used for curve-fitting and deconvolution, and smart background subtraction. XRD measurements were obtained using a Bruker-AXS D8-Advance diffractometer with Cu K α 1.54 Å radiation.

The chemical composition was determined by organic elemental analysis UNICUBE by Elementar and ICP using SPECTROARCOS ICP OES, FHX22, MultiView plasma. For ICP analysis, aerogels samples of 2-3 mg were heated to 500 °C for 20 h in glass test tubes, and dissolved in 32% HCl, that was then diluted with deionized water.

Structural characterization was performed by gas adsorption analysis, using Quantachrome Autosorb iQ instrument with nitrogen as the adsorbate at 77 K. Surface area was calculated from the BET equation. The pore size distribution was calculated using the DFT method by N₂ at 77 K on carbon: slit/cylindrical or slit pore, NLDFT equilibrium model, that is recommended for microporous structure and enable the best fit to the BET surface area. The slit/cylindrical pore model was used for PTEB₂bpyFe-HT2, PTEB₁bpyFe-HT2, and PTEB_{0.33}bpyFe-HT2. The slit pore model was used

for PTEB₃bpyFe-HT2 and PTEB₄bpyFe-HT2. SE- and ADF-STEM images were recorded on a probe-corrected JEOL NEOARM operated at 80 kV and equipped with a cold field emission gun. Raman spectroscopy measurements were performed using DXR1, with DXR 532 nm laser and high-resolution grating. Measurement conditions: enlargement ×10, spectrograph aperture 25 μm pinhole, laser power level 7 mW.

The electrochemical studies were performed using a Bio-Logic VMP 300 bipotentiostat and Pine MSR Electrode Rotator. The catalysts' slurries were prepared by grinding the aerogel and forming a suspension in isopropanol: deionized water (2:1 v/v) containing 0.2 % wt Nafion at a concentration of 10 mg aerogel per ml. For CV and RRDE measurements the slurry (6.3 μL) was drop-cast on a glassy carbon (GC) disk of a RRDE electrode (Pine, 5.61 mm diameter, with Platinum ring). The measurements were conducted in 0.1 M KOH (Acros, 99.98%) and 0.5 M H₂SO₄ aqueous solutions, using a 3 mm diameter GC rod as a counter electrode and a RHE as a reference in a Teflon cell. O₂ (99.999%) was purged in the solution to test the electrocatalytic performance of the aerogels, and pure Ar (99.999%) was purged for background current measurements. CV measurements were performed at a scan rate of 100 mV/s. RRDE measurements were performed at a scan rate of 5 mV/s and rotational speed of 900 rpm. The aerogels were compared to commercial Fe-based PGM-free catalysts (Pajarito Powder PMF-012101).

Supporting Information

Additional characterizations are available in the supporting information, including STEM, ICP, XPS, Raman, N₂ adsorption, XRD and electrochemical measurements. The electron number, loading by ICP, EASD and TOF calculations are also explained.

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References

1. Lowitzsch, J.; Hoicka, C. E.; van Tulder, F. J., Renewable energy communities under the 2019 European Clean Energy Package—

Governance model for the energy clusters of the future? *Renewable and Sustainable Energy Reviews* **2020**, *122*, 109489.

2. Farrauto, R. J.; Deeba, M.; Alerasool, S., Gasoline automobile catalysis and its historical journey to cleaner air. *Nature Catalysis* **2019**, *2* (7), 603-613.
3. Kovač, A.; Paranos, M.; Marciuš, D., Hydrogen in energy transition: A review. *International Journal of Hydrogen Energy* **2021**, *46* (16), 10016-10035.
4. Jaouen, F.; Jones, D.; Coutard, N.; Artero, V.; Strasser, P.; Kucernak, A., Toward platinum group metal-free catalysts for hydrogen/air proton-exchange membrane fuel cells. *Johnson Matthey Technology Review* **2018**, *62* (2), 231-255.
5. Jiao, K.; Xuan, J.; Du, Q.; Bao, Z.; Xie, B.; Wang, B.; Zhao, Y.; Fan, L.; Wang, H.; Hou, Z., Designing the next generation of proton-exchange membrane fuel cells. *Nature* **2021**, *595* (7867), 361-369.
6. Bashyam, R.; Zelenay, P., A class of non-precious metal composite catalysts for fuel cells. *Nature* **2006**, *443* (7107), 63-66.
7. Kaur, M.; Pal, K., Review on hydrogen storage materials and methods from an electrochemical viewpoint. *Journal of Energy Storage* **2019**, *23*, 234-249.
8. Tseng, P.; Lee, J.; Friley, P., A hydrogen economy: opportunities and challenges. *Energy* **2005**, *30* (14), 2703-2720.
9. Qin, D.-D.; Tang, Y.; Ma, G.; Qin, L.; Tao, C.-L.; Zhang, X.; Tang, Z., Molecular metal nanoclusters for ORR, HER and OER: Achievements, opportunities and challenges. *International Journal of Hydrogen Energy* **2021**, *46* (51), 25771-25781.
10. Shen, L.; Ma, M.; Tu, F.; Zhao, Z.; Xia, Y.; Goh, K.; Zhao, L.; Wang, Z.; Shao, G., Recent advances in high-loading catalysts for low-temperature fuel cells: from nanoparticle to single atom. *SusMat* **2021**, *1* (4), 569-592.
11. Gasteiger, H. A.; Marković, N. M., Just a dream—or future reality? *science* **2009**, *324* (5923), 48-49.
12. Wan, C.; Duan, X.; Huang, Y., Molecular design of single-atom catalysts for oxygen reduction reaction. *Advanced Energy Materials* **2020**, *10* (14), 1903815.
13. Thompson, S. T.; Papageorgopoulos, D., Platinum group metal-free catalysts boost cost competitiveness of fuel cell vehicles. *Nature Catalysis* **2019**, *2* (7), 558-561.

14. He, Y.; Wu, G., PGM-Free Oxygen-Reduction Catalyst Development for Proton-Exchange Membrane Fuel Cells: Challenges, Solutions, and Promises. *Accounts of Materials Research* **2022**, *3* (2), 224-236.
15. Hossen, M. M.; Hasan, M. S.; Sardar, M. R. I.; bin Haider, J.; Tammeveski, K.; Atanassov, P., State-of-the-art and developmental trends in platinum group metal-free cathode catalyst for anion exchange membrane fuel cell (AEMFC). *Applied Catalysis B: Environmental* **2022**, 121733.
16. Martinez, U.; Komini Babu, S.; Holby, E. F.; Chung, H. T.; Yin, X.; Zelenay, P., Progress in the development of Fe-based PGM-free electrocatalysts for the oxygen reduction reaction. *Advanced materials* **2019**, *31* (31), 1806545.
17. Bates, J. S.; Biswas, S.; Suh, S.-E.; Johnson, M. R.; Mondal, B.; Root, T. W.; Stahl, S. S., Chemical and electrochemical O₂ reduction on earth-abundant MNC catalysts and implications for mediated electrolysis. *Journal of the American Chemical Society* **2022**, *144* (2), 922-927.
18. Zion, N.; Friedman, A.; Levy, N.; Elbaz, L., Bioinspired Electrocatalysis of Oxygen Reduction Reaction in Fuel Cells Using Molecular Catalysts. *Adv Mater* **2018**, *30*, 1800406.
19. Guan, Y.; Lai, J.; Xu, G., Recent Advances on Electrocatalysis Using Pristine Conductive Metal-Organic Frameworks and Covalent Organic Frameworks. *ChemElectroChem* **2021**, *8* (15), 2764-2777.
20. Cai, B.; Eychmüller, A., Promoting electrocatalysis upon aerogels. *Advanced Materials* **2019**, *31* (31), 1804881.
21. Peles-Strahl, L.; Persky, Y.; Elbaz, L., Design of advanced aerogel structures for oxygen reduction reaction electrocatalysis. *SusMat* **2023**, *3* (1), 44-57.
22. Peles-Strahl, L.; Zion, N.; Lori, O.; Levy, N.; Bar, G.; Dahan, A.; Elbaz, L., Bipyridine Modified Conjugated Carbon Aerogels as a Platform for the Electrocatalysis of Oxygen Reduction Reaction. *Advanced Functional Materials* **2021**, 2100163.
23. Zion, N.; Cullen, D. A.; Zelenay, P.; Elbaz, L., Heat-Treated Aerogel as a Catalyst for the Oxygen Reduction Reaction. *Angew Chem Int Ed Engl* **2020**, *59* (6), 2483-2489.
24. Cai, B.; Hübner, R.; Sasaki, K.; Zhang, Y.; Su, D.; Ziegler, C.; Vukmirovic, M. B.; Rellinghaus, B.; Adzic, R. R.; Eychmüller, A., Core-shell structuring of pure metallic aerogels towards highly efficient platinum utilization for the oxygen reduction reaction. *Angewandte Chemie International Edition* **2018**, *57* (11), 2963-2966.
25. Zhang, X.; Wang, C.; Chen, K.; Clark, A. H.; Hübner, R.; Zhan, J.; Zhang, L.; Eychmüller, A.; Cai, B., Optimizing the Pd sites in pure metallic aerogels for efficient electrocatalytic H₂O₂ production. *Advanced Materials* **2023**, *35* (14), 2211512.
26. Zion, N.; Peles-Strahl, L.; Friedman, A.; Cullen, D. A.; Elbaz, L., Electrocatalysis of oxygen reduction reaction in a polymer electrolyte fuel cell with a covalent framework of iron phthalocyanine aerogel. *ACS Applied Energy Materials* **2022**, *5* (7), 7997-8003.
27. Long, S.; Feng, Y.; He, F.; He, S.; Hong, H.; Yang, X.; Zheng, L.; Liu, J.; Gan, L.; Long, M., An ultralight, supercompressible, superhydrophobic and multifunctional carbon aerogel with a specially designed structure. *Carbon* **2020**, *158*, 137-145.
28. Bezerra, C. W.; Zhang, L.; Liu, H.; Lee, K.; Marques, A. L.; Marques, E. P.; Wang, H.; Zhang, J., A review of heat-treatment effects on activity and stability of PEM fuel cell catalysts for oxygen reduction reaction. *Journal of Power Sources* **2007**, *173* (2), 891-908.
29. He, Y.; Tan, Q.; Lu, L.; Sokolowski, J.; Wu, G., Metal-nitrogen-carbon catalysts for oxygen reduction in PEM fuel cells: self-template synthesis approach to enhancing catalytic activity and stability. *Electrochemical Energy Reviews* **2019**, *2* (2), 231-251.
30. Levy, N.; Mahammed, A.; Friedman, A.; Gavriel, B.; Gross, Z.; Elbaz, L., Metalloporphyrins as Non-Precious Metal Electrocatalysts for Highly Efficient Oxygen Reduction in Alkaline Media. *ChemCatChem* **2016**, *8* (17), 2832-2837.
31. Zion, N.; Douglin, J. C.; Cullen, D. A.; Zelenay, P.; Dekel, D. R.; Elbaz, L., Porphyrin aerogel catalysts for oxygen reduction reaction in anion-exchange membrane fuel cells. *Advanced Functional Materials* **2021**, *31* (24), 2100963.
32. Wu, G.; Johnston, C. M.; Mack, N. H.; Artyushkova, K.; Ferrandon, M.; Nelson, M.; Lezama-Pacheco, J. S.; Conradson, S. D.; More, K. L.; Myers, D. J., Synthesis-structure-performance correlation for polyaniline-Me-C non-precious metal cathode catalysts for oxygen reduction in

- fuel cells. *Journal of Materials Chemistry* **2011**, 21 (30), 11392-11405.
33. Chen, Y.; Liu, Y.; Sun, C.; Yu, K.; Song, M.; Wang, H.; Zhang, X., Microstructure and strengthening mechanisms in Cu/Fe multilayers. *Acta Materialia* **2012**, 60 (18), 6312-6321.
 34. Lu, E.; Wu, J.; Yang, B.; Yu, D.; Yu, Z.; Hou, Y.; Zhang, J., Selective hydroxylation of benzene to phenol over Fe nanoparticles encapsulated within N-doped carbon shells. *ACS Applied Nano Materials* **2020**, 3 (9), 9192-9199.
 35. Omar, H. D., The analysis of copper-iron metallic mixture by means of XRD and XRF. *International Letters of Chemistry, Physics and Astronomy* **2016**, 64, 130-134.
 36. Matanovic, I.; Artyushkova, K.; Strand, M. B.; Dzara, M. J.; Pylypenko, S.; Atanassov, P., Core level shifts of hydrogenated pyridinic and pyrrolic nitrogen in the nitrogen-containing graphene-based electrocatalysts: in-plane vs edge defects. *The Journal of Physical Chemistry C* **2016**, 120 (51), 29225-29232.
 37. Artyushkova, K., Misconceptions in interpretation of nitrogen chemistry from x-ray photoelectron spectra. *Journal of Vacuum Science & Technology A: Vacuum, Surfaces, and Films* **2020**, 38 (3), 031002.
 38. Marshall-Roth, T.; Libretto, N. J.; Wrobel, A. T.; Anderton, K. J.; Pegis, M. L.; Rieke, N. D.; Voorhis, T. V.; Miller, J. T.; Surendranath, Y., A pyridinic Fe-N₄ macrocycle models the active sites in Fe/N-doped carbon electrocatalysts. *Nature communications* **2020**, 11 (1), 1-14.
 39. Friedman, A.; Mizrahi, M.; Levy, N.; Zion, N.; Zachman, M.; Elbaz, L., Application of Molecular Catalysts for the Oxygen Reduction Reaction in Alkaline Fuel Cells. *ACS Applied Materials & Interfaces* **2021**, 13 (49), 58532-58538.
 40. Levy, N.; Mahammed, A.; Kosa, M.; Major, D. T.; Gross, Z.; Elbaz, L., Metalloporphyrins as Nonprecious-Metal Catalysts for Oxygen Reduction. *Angewandte Chemie International Edition* **2015**, 54 (47), 14080-14084.
 41. Sengupta, K.; Chatterjee, S.; Dey, A., In situ mechanistic investigation of O₂ reduction by iron porphyrin electrocatalysts using surface-enhanced resonance Raman spectroscopy coupled to rotating disk electrode (SERRS-RDE) setup. *ACS Catalysis* **2016**, 6 (10), 6838-6852.
 42. Pegis, M. L.; McKeown, B. A.; Kumar, N.; Lang, K.; Wasylenko, D. J.; Zhang, X. P.; Raugei, S.; Mayer, J. M., Homogenous electrocatalytic oxygen reduction rates correlate with reaction overpotential in acidic organic solutions. *ACS Central Science* **2016**, 2 (11), 850-856.
 43. Snitkoff-Sol, R. Z.; Elbaz, L., Assessing and measuring the active site density of PGM-free ORR catalysts. *Journal of Solid State Electrochemistry* **2022**, 26 (9), 1839-1850.
 44. Kozhushner, A.; Zion, N.; Elbaz, L., Methods for assessment and measurement of the active site density in platinum group metal-free oxygen reduction reaction catalysts. *Current Opinion in Electrochemistry* **2021**, 25, 100620.
 45. Specchia, S.; Atanassov, P.; Zagal, J. H., Mapping transition metal-nitrogen-carbon catalyst performance on the critical descriptor diagram. *Current Opinion in Electrochemistry* **2021**, 27, 100687.
 46. Malko, D.; Kucernak, A.; Lopes, T., In situ electrochemical quantification of active sites in Fe-N/C non-precious metal catalysts. *Nature Communications* **2016**, 7, 13285.
 47. Luo, F.; Choi, C. H.; Primbs, M. J. M.; Ju, W.; Li, S.; Leonard, N. D.; Thomas, A.; Jaouen, F.; Strasser, P., Accurate Evaluation of Active-Site Density (SD) and Turnover Frequency (TOF) of PGM-Free Metal-Nitrogen-Doped Carbon (MNC) Electrocatalysts using CO Cryo Adsorption. *ACS Catalysis* **2019**, 9 (6), 4841-4852.
 48. Friedman, A.; Snitkoff-Sol, R. Z.; Honig, H. C.; Elbaz, L., Simplified FTacV model to quantify the electrochemically active site density in PGM-free ORR catalysts. *Electrochimica Acta* **2023**, 463, 142865.
 49. Snitkoff-Sol, R. Z.; Friedman, A.; Honig, H. C.; Yurko, Y.; Kozhushner, A.; Zachman, M. J.; Zelenay, P.; Bond, A. M.; Elbaz, L., Quantifying the electrochemical active site density of precious metal-free catalysts in situ in fuel cells. *Nature Catalysis* **2022**, 5 (2), 163-170.
 50. Kozhushner, A.; Zion, N.; Elbaz, L., Methods for Assessment and Measurement of the Active Site Density in PGM-free ORR Catalysts. *Current Opinion in Electrochemistry* **2020**, 25, 100620.

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